

CAIRO REAL ESTATE

Market Analysis

How Location & Property Type Drive Prices
Across 28 Districts in Cairo

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BY THE NUMBERS

Dataset at a Glance

11,418

Raw Listings

From Egyptian platforms

8,851

Clean Records

After IQR outlier removal

28

Districts

Across Cairo

10

Property Types

Incl. villas, apartments & more

What's Driving Prices in Cairo?

Context

Cairo's housing market is changing quickly because the city is growing fast and more people need homes.

Prices can be very different depending on the area and the type of property, so buyers and investors should rely on data rather than just guessing.

This study analyzes 8,851 cleaned listings scraped from Egyptian real estate platforms to statistically determine what actually drives price per square meter.

Research Question

Do location and property type significantly affect price per sqm?



District / Location

28 Cairo districts analyzed



Property Type

Apartments, Villas, Twin Houses...

How We Built the Analysis

01

Data Collection

11,418 Kaggle listings from Egyptian real estate platforms (Spring 2026)

02

Cleaning

Removed 412 'Ask' entries · IQR-based outlier removal on price & size_sqm

03

Feature Engineering

Parsed location → city / district / compound · Engineered price_per_sqm & total_rooms

04

Statistical Tests

One-way ANOVA · Kruskal-Wallis · OLS regression on log-transformed price

HYPOTHESIS

What We Set Out to Test

H_0 Null Hypothesis

Location and property type have NO significant effect on price per sqm.

Any observed differences are due to random variation, not systematic factors.

H_1 Alternative Hypothesis

At least one factor district or property type significantly predicts price per sqm.

Location and type create statistically distinct price groups that go beyond random chance.

Significance level: $\alpha = 0.05$ · Both parametric (ANOVA) and non-parametric (Kruskal-Wallis) tests applied for robustness

STATISTICAL RESULTS

All 4 Tests Reject H_0

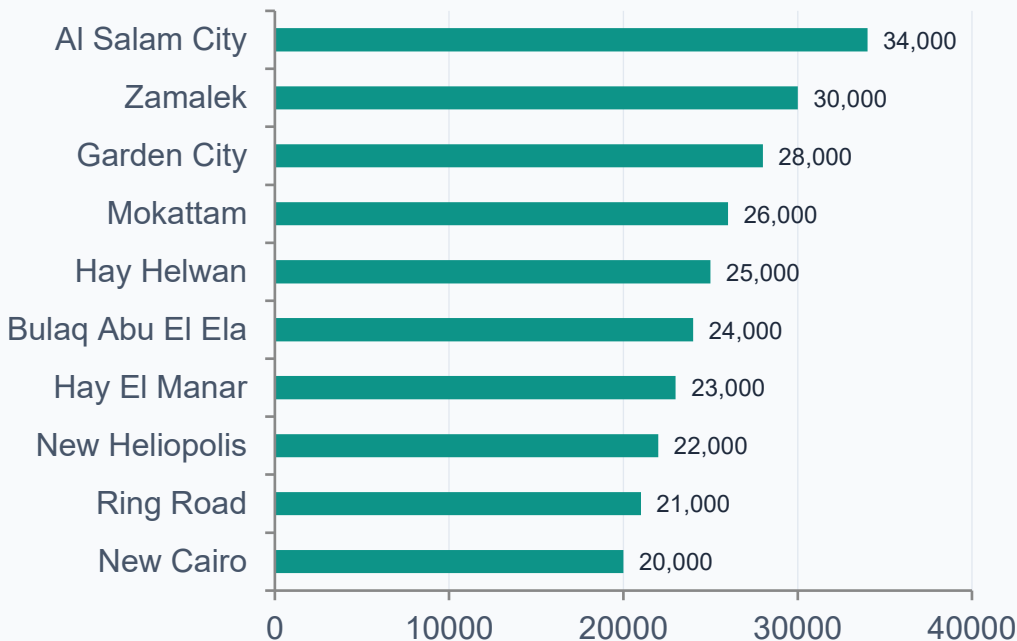
Test	Factor	p-value	Result
ANOVA	Property Type vs Price/sqm	< 0.05	✓ Reject H_0
ANOVA	District vs Price/sqm	< 0.05	✓ Reject H_0
Kruskal-Wallis	Property Type vs Price/sqm	< 0.05	✓ Reject H_0
Kruskal-Wallis	District vs Price/sqm	< 0.05	✓ Reject H_0

Both parametric and non-parametric approaches confirm the same conclusion results are robust to distributional assumptions.

KEY FINDINGS

What the Data Reveals

Top 10 Districts by Avg Price/sqm (EGP)



 **34,000 EGP/sqm**

Al Salam City — top district

 **+70%**

Al Salam vs New Cairo gap

 **2×**

Villas vs apartments premium

 **$r = 0.93$**

Bedrooms & bathrooms correlation

Type Matters as Much as Location



Villas & Twin Houses

Command 2× the price of a typical apartment
buyers pay a large premium beyond just square
meters.

Size ≠ Price

Size alone is NOT a strong predictor. Even at
similar sizes, property type creates distinct price
tiers.

Modelling Tip

Log-transform essential for regression. Use
total_rooms to avoid multicollinearity (bedrooms
 $r = 0.93$ with bathrooms).

What This Means in Practice



For Investors

- Target under-priced districts before prices converge
- Al Salam City leads but watch for growth in Ring Road & New Heliopolis
- Type + location beats size for pricing decisions



For Policy

- Build more apartments to make housing more affordable.
- The big price gap between villas and apartments shows a growing wealth divide.
- Use district-level price data to guide city planning decisions.



For Analysts

- Always log-transform price before regression
- ANOVA & Kruskal-Wallis both confirm findings are robust
- Use total_rooms to avoid bedroom/bathroom multicollinearity

CONCLUSION

The Verdict

$p < 0.05$
across all 4 tests

H_0 Rejected

DATA DECODERS TEAM



Both district and property type are statistically significant price predictors.



Location alone can swing prices 50%+ above average — Al Salam City leads at ~34,000 EGP/sqm.



Villas cost 2× more than apartments even after controlling for size.



Size alone is insufficient type + location > size for pricing decisions.